Classified

Geotopia Police Department

Violation Record

Access to this document is restricted to individuals with the proper security clearance and need-to-know. Violations are subject to severe penalties under applicable laws.

Geotopia Police Department
Violation Record

Name:
Jay Gemini

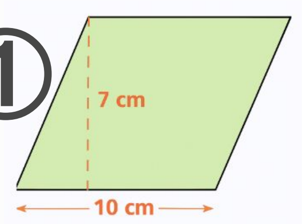
Address:
145 Tangram St.
Geotopia, GT 11223

ID Number:
3062471571

Photo:

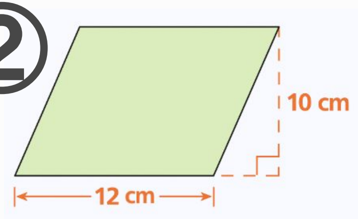
Details of Violation:

①



Area of it must be 17!
10+7=17 cm

②



Area is 120cm?

Officer's note:

Area of Parallelogram

$$\boxed{A_{\square} = \text{base} \cdot \text{height}}$$

① Base = 10cm Height = 7cm

$$A_{\square} = 10 \cdot 7 = 70\ cm^2$$

② Base = 12cm Height = 10cm

$$A_{\square} = 12 \cdot 10 = 120\ cm^2$$

Geotopia Police Department
Violation Record

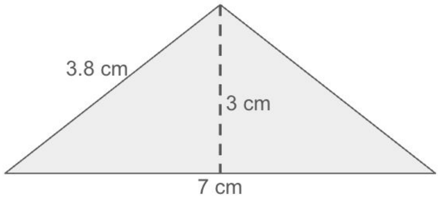
Name:
Kami James

Address:
831 Tangram St.
Geotopia, GT 11223

ID Number:
9489169881

Photo:

Details of Violation:



Area:
3 x 7 = 21 cm

Officer's note: Area of Triangle

$$\boxed{A_\triangle = Base \cdot Height \div 2}$$

$Area_{\square} = 5 \times 6 = 30\ cm^2$

$Area_\triangle = 5 \cdot 6 \div 2 = 15\ cm^2$

Base = 7 Height = 3

$$Area_\triangle = 7 \cdot 3 \div 2 = 10.5\ cm^2$$

Area of △ is half of base times height!

Geotopia Police Department
Violation Record

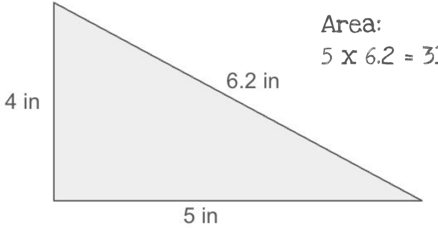
Name:
Jack Marion

Address:
781 Union St.
Geotopia, GT 01301

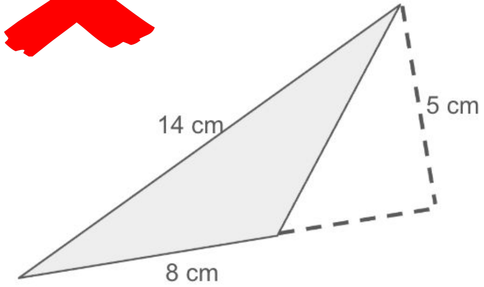
ID Number:
7658118390

Photo:

Details of Violation:



Area:
5 x 6.2 = 31



Area: 8 x 5 = 40 cm

Officer's note:

Base = 4 Height = 5

$$Area_{\triangle} = 4 \cdot 5 \div 2 = 10 \text{ in}^2$$

Base = 8 Height = 5

$$Area_{\triangle} = 8 \cdot 5 \div 2 = 20 \text{ cm}^2$$

Geotopia Police Department
Violation Record

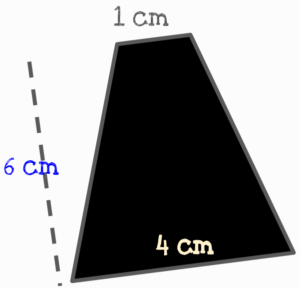
Name:
Victor Li

Address:
26 Chopstic St.
Geotopia, GT 11249

ID Number:
7745641289

Photo:

Details of Violation:



What is this?

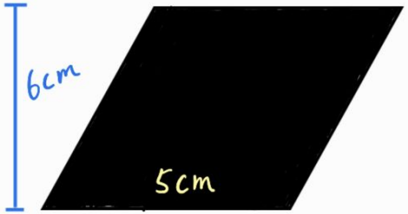
Is it a rectangle?

So the area must be 9x10=90!

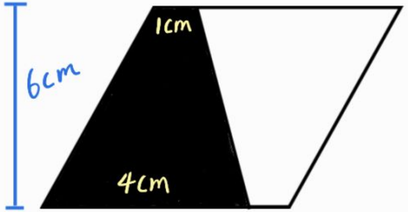
Officer's note:

Area of Trapezoid

$$\boxed{A_\triangle = (\text{base } 1 + \text{base } 2) \cdot \text{height} \div 2}$$



$A_\square = 6 \cdot 5 = 30 \text{ cm}^2$



$A_\triangle = 30 \div 2 = 15 \text{ cm}^2$

$= (1+4) \cdot 6 \div 2$

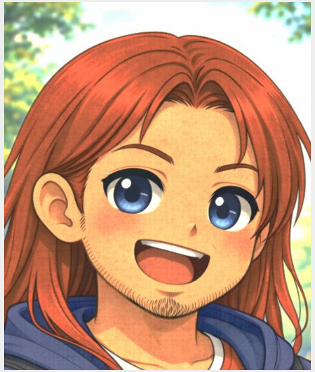
Geotopia Police Department
Violation Record

Name:
George Wiseman

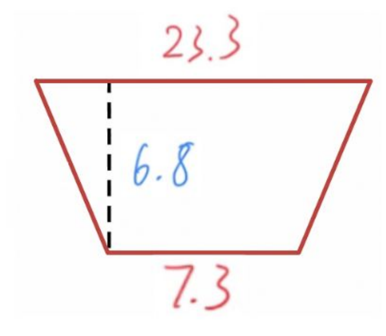
Address:
72 Chopstic St.
Geotopia, GT 11249

ID Number:
7745641300

Photo:



Details of Violation:



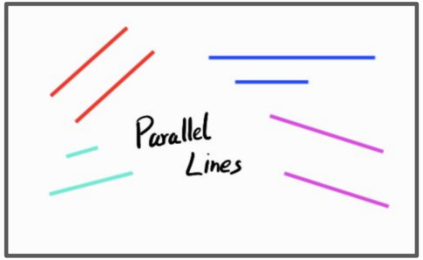
Area:
Bases are 7.3 and 6.8
Height is 23.3

So area is 7.3 x 6.8 x 23.3

$$A_{\triangle} = (\text{base } 1 + \text{base } 2) \cdot \text{height} \div 2$$

Officer's note:

How to find base 1 and 2?
They are parallel lines!



Base 1 = 23.3
Base 2 = 7.3
} Parallel

Height = 6.8

$$\text{Area}_{\triangle} = (23.3 + 7.3) \cdot 6.8 \div 2$$
$$= 104.04 \text{ in}^2$$

Geotopia Police Department
Violation Record

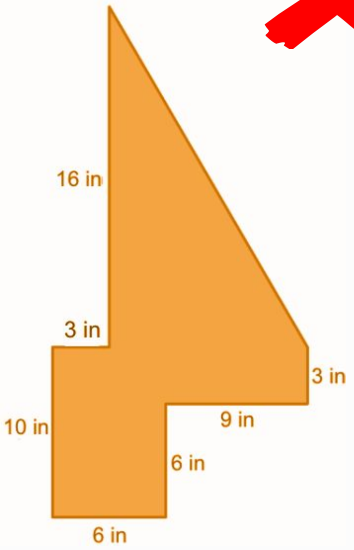
Name:
Mario Jaquez

Address:
381 Res Pl.
Geotopia, GT 05671

ID Number:
8955077691

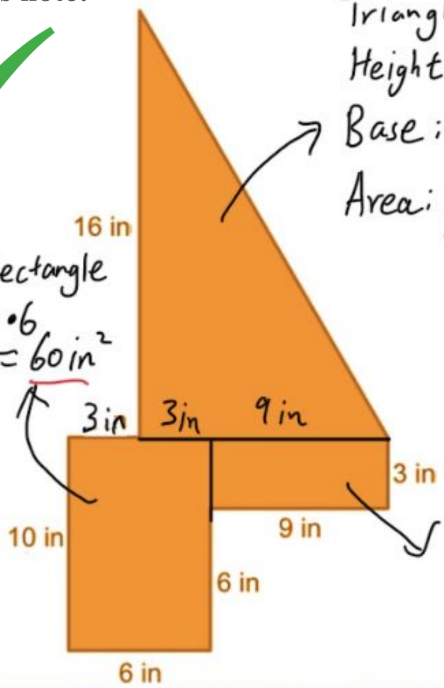
Photo:

Details of Violation:



Area:
16+3+10+6+6+6+9+3=62

Officer's note:



Triangle:
Height: 16 in
Base: $3+9=12$ in
Area: $\frac{1}{2}\cdot 16\cdot 12 = 96\,in^2$

Rectangle
$10\cdot 6$
$= 60\,in^2$

Rectangle
$9\cdot 3 = 27\,in^2$

Total Area: $96+60+27$
$= 183\,in^2$

Geopotia Police Department
Violation Record

Name:
Joey Inaibbirt

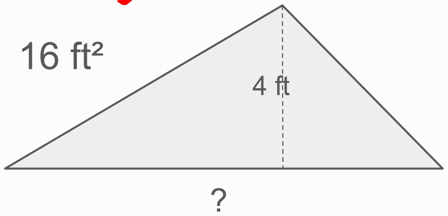
Address:
90 Bedford st.
Geotopia, GT 10014

ID Number:
3141592653

Photo:

Details of Violation:

The missing height is 5ft!



The missing base is 4ft!

Officer's note:

$A_{\square} = \text{base} \cdot \text{height}$

$22.5 = 5 \cdot \text{height}$

$\div 5 \qquad \div 5$

$\boxed{4.5 = \text{height}}$

The height is 4.5 ft

$A_{\triangle} = \text{base} \cdot \text{height} \div 2$

$16 = \text{base} \cdot 4 \div 2$

$16 = \text{base} \cdot 2$

$\boxed{8 = \text{base}}$

The base is 8 ft

Geotopia Police Department
Violation Record

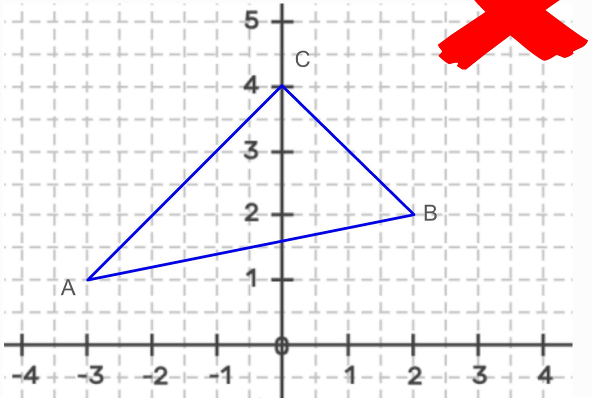
Name:
Annie Haywood

Address:
101 Campus Road.
Geotopia, GT 18990

ID Number:
4516677180

Photo:

Details of Violation:

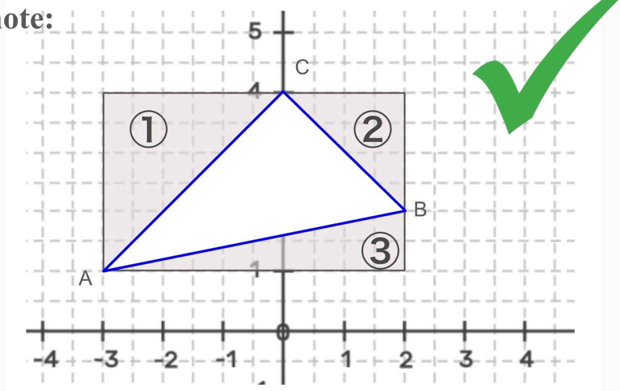


What is the area of this triangle?

I see the base is 4 and the height is 5 so the area is 4x5/2=10

We can't find the base or height of that triangle, but we can do:

Officer's note:



= Whole − 1 − 2 − 3

Whole	1	2	3
Length = 5 (-3 to 2)	Base = 3 (-3 to 0)	Base = 2	Base = 5
Width = 3 (1 to 4)	Height = 3 (1 to 4)	Height = 2	Height = 1
$A_{\square} = 5 \cdot 3 = 15$	$A_{\triangle} = 3 \cdot 3 \div 2 = 4.5$	$A_{\triangle} = 2 \cdot 2 \div 2 = 2$	$A_{\triangle} = 5 \cdot 1 \div 2 = 2.5$

$$6 = 15 - 4.5 - 2 - 2.5$$

Geotopia Police Department
Violation Record

Name:
Kaya Noshu

Address:
301 Edaw st.
Geotopia, GT 15400

ID Number:
4499177188

Photo:

Details of Violation:

What is the perimeter of the rectangle with the vertices of (5, -1), (5, 3), (0, -1) and (0, 3)?

I guess it is 7?

Officer's note:

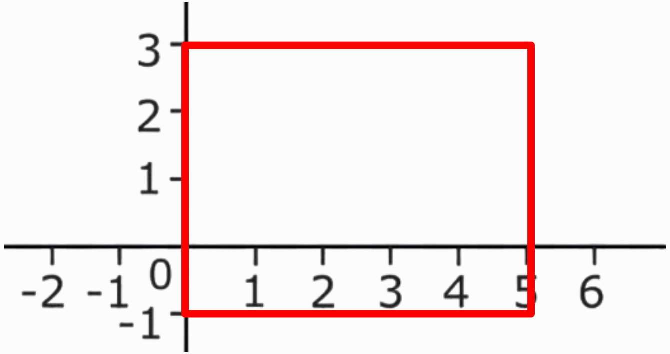
There are 4 units from (5, -1) to (5, 3)

There are 5 units from (5, 3) to (0, 3)

There are 4 units from (0, 3) to (0, -1)

There are 5 units from (5, -1) to (0, -1)

So 4+5+4+5=18 is the perimeter!



Geotopia Police Department
Violation Record

Name:
Jose Henderson

Address:
303 Edaw st.
Geotopia, GT 15400

ID Number:
4499177183

Photo:

Details of Violation:

I guess the perfect squares are 2, 3, 4 5...

I guess the perfect cubes are perfect ice cubes...

Officer's note:


4


9


16

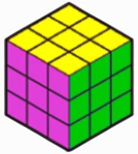
1x1=1 2x2=4

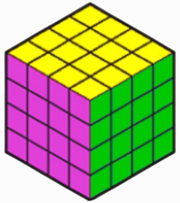
3x3=9 4x4=16

5x5=25 6x6=36

49, 64, 81...these numbers are perfect squares.


8


27


64

1x1x1=1 2x2x2=8

3x3x3=27 4x4x4=64

125, 216, 343...these numbers are perfect cubes.

Geotopia Police Department
Violation Record

Name:
Airik Adams

Address:
666 City Hall.
Geotopia, GT 10001

ID Number:
3151322707

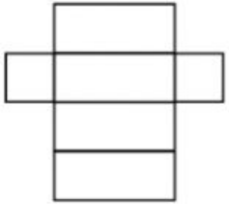
Photo:

Details of Violation:

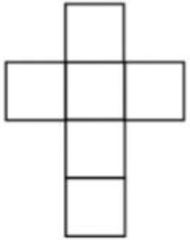
What are they?

They are boxes!

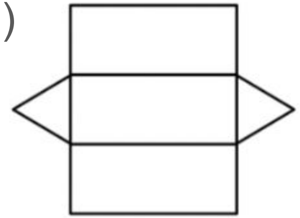
1)



2)



3)

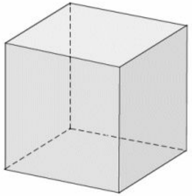


Officer's note:

This is a rectangular prism:

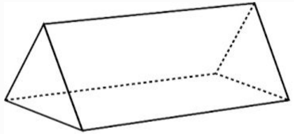
- 8 vertices (corners)
- 12 edges (sides)
- 6 faces

This is a special rectangular prism known as a cube.



This is a triangular prism:

- 6 vertices
- 9 edges
- 5 faces



Geotopia Police Department
Violation Record

Name:
Alejandro García

Address:
89 Volume Avenue
Geotopia, GT 12341

ID Number:
0678833422

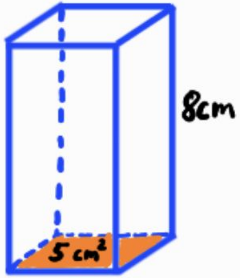
Photo:

Details of Violation: I have a rectangular prism with 2 in, 3 in and 10 in as the length, width and height.

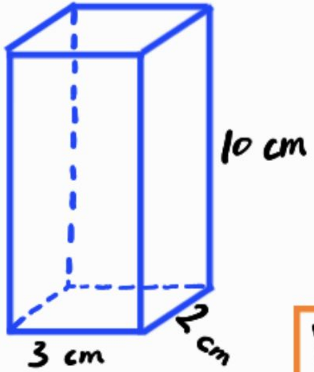
It's volume is 3+2+10=15!

Officer's note:

How to find Volume



$$V = \text{Base Area} \cdot \text{Height}$$
$$= 5 \cdot 8$$
$$= 40 \text{ cm}^3$$



$$V = \text{Base Area} \cdot \text{Height}$$
$$= \text{Length} \cdot \text{Width} \cdot \text{Height}$$
$$= 3 \cdot 2 \cdot 10$$
$$= 60 \text{ cm}^3$$

$$\boxed{V = l \cdot w \cdot h}$$

Geotopia Police Department
Violation Record

Name:
Aahana Gupta

Address:
47 Surface Avenue
Geotopia, GT 10011

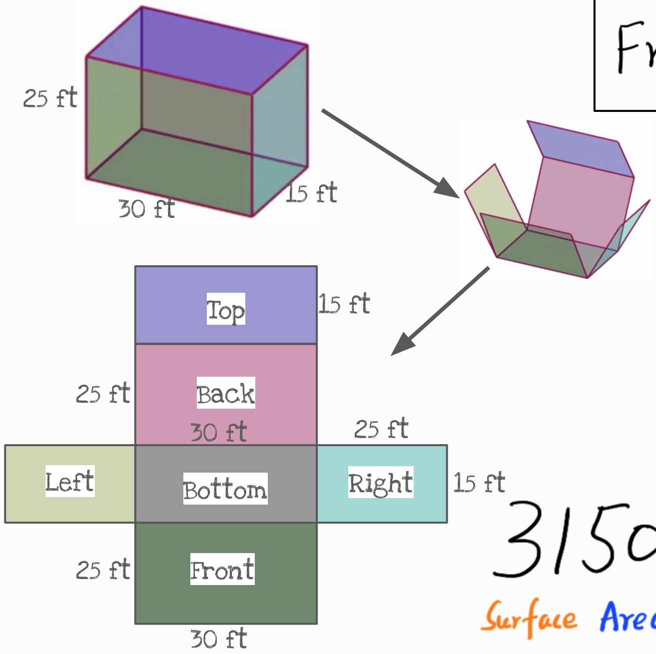
ID Number:
1238567723

Photo:

Details of Violation: I know the surface area of this rectangular prism which has length, width and height in 30 ft, 15 ft and 25 ft.

It's volume is 30+25+15=70!

Officer's note:



Surface Area of Rectangular Prism

$\underbrace{\text{Front} + \text{Back}}_{\text{Same}} + \underbrace{\text{Top} + \text{Bottom}}_{\text{Same}} + \underbrace{\text{Left} + \text{Right}}_{\text{Same}}$

Shortcut

$$\boxed{SA = (\text{Front} + \text{Bottom} + \text{Right}) \cdot 2}$$

$30 \cdot 25 \quad 30 \cdot 15 \quad 25 \cdot 15$

$$3150\ ft^2 = (750 + 450 + 375) \cdot 2$$

Surface Area

Geotopia Police Department
Violation Record

Name:
Stephenie Anderson

Address:
135 Prism Avenue
Geotopia, GT 10323

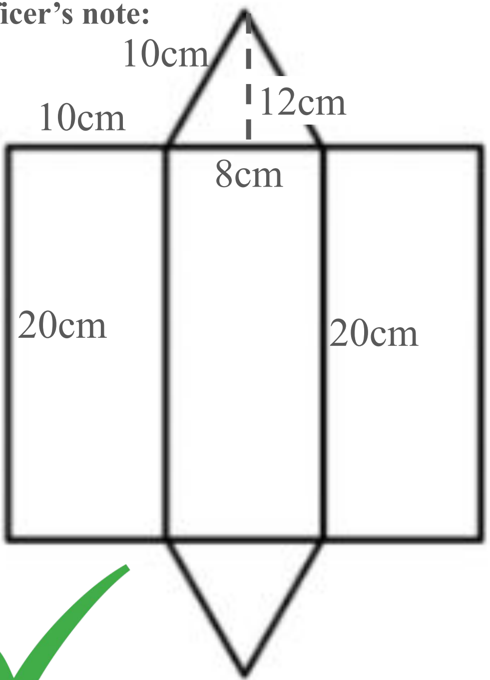
ID Number:
1456119973

Photo:

Details of Violation: I know the surface area of triangular prism which has base of 8 cm, height of 12 cm, length of 20 cm and slant height of 10 cm

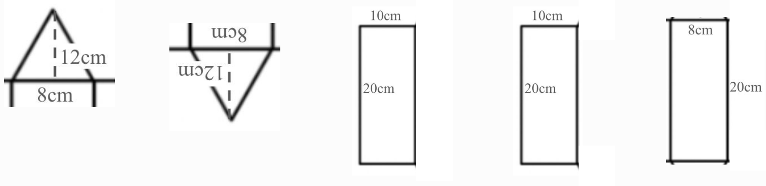
It's volume is 8+12+20+10=60!

Officer's note:



Surface Area:

Front + Back+Right+Left+Bottom



½*8*12 + ½*8*12 + 20*10 + 20*10 + 8*20

= **656** cm²